

QAARSUT UUMMANNAQ, Greenland

WMO: 042130

Lat: 70.733N

Lon: 52.700W

Elev: 292

StdP: 14.54

Time Zone: -3.00 (GRL)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-16.0	-12.0	-28.7	1.1	-13.3	-24.0	1.5	-9.0	35.0	16.5	31.3	16.6	6.8	140	0.958

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	61.0	46.2	58.3	45.3	56.0	44.7	49.2	53.1	47.9	52.4	46.9	52.1	19.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
47.9	49.9	49.0	46.5	47.2	47.9	45.0	44.6	46.9	19.9	52.9	19.3	52.3	18.7	52.1	53.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.9	28.6	25.5	DB	-16.7	68.0	6.0	9.3	-21.0	74.7	-24.4	80.1	-27.8	85.4	-32.1	92.2	(4)
(5)			WB	-16.6	51.2	6.1	1.3	-21.0	52.2	-24.5	52.9	-27.9	53.7	-32.3	54.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.9	10.8	7.6	7.6	17.7	31.2	42.9	48.8	46.2	36.5	26.6	19.6	14.3	(6)
(7)		DBStd	16.87	10.60	12.71	12.09	11.27	7.42	6.28	5.63	4.45	5.15	7.05	8.32	8.89	(7)
(8)		HDD50	8869	1215	1186	1315	972	582	226	90	133	404	726	912	1108	(8)
(9)		HDD65	14262	1680	1606	1780	1421	1046	664	502	583	854	1191	1362	1573	(9)
(10)		CDD50	84	0	0	0	2	1	12	54	15	0	0	0	0	(10)
(11)		CDD65	1	0	0	0	1	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	23	0	0	0	23	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	10	0	0	0	10	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	10.3	11.5	10.4	8.2	8.4	9.1	9.6	9.8	9.7	9.8	11.1	12.9	12.6	(14)
(15)	Precipitation	PrecAvg	17.0	1.5	1.0	0.9	0.9	0.9	0.9	1.1	1.8	1.7	1.9	2.1	2.1	(15)
(16)		PrecMax	25.2	5.6	2.2	3.0	2.7	2.7	2.7	2.8	3.8	3.9	4.4	5.4	4.3	(16)
(17)		PrecMin	8.7	0.1	0.1	0.1	0.0	0.1	0.1	0.1	0.6	0.4	0.5	0.2	0.3	(17)
(18)		PrecStd	4.2	1.1	0.7	0.7	0.6	0.6	0.6	0.6	0.9	0.8	1.0	1.1	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	41.7	41.5	73.9	56.4	61.4	64.8	61.7	51.0	47.7	44.7	40.2	(19)
(20)			MCWB	33.4	32.7	35.2	47.2	43.4	45.6	48.1	46.8	40.1	37.3	37.3	33.1	(20)
(21)		2%	DB	34.0	33.9	34.4	45.6	49.7	58.2	62.1	57.7	48.4	43.3	40.0	34.4	(21)
(22)			MCWB	28.6	26.9	29.7	37.3	40.7	44.9	46.9	44.4	39.4	34.8	33.1	29.8	(22)
(23)		5%	DB	30.4	30.6	29.1	38.8	45.5	55.7	59.6	55.6	45.8	39.2	36.6	30.3	(23)
(24)			MCWB	25.8	25.4	24.9	32.2	38.2	44.2	45.9	44.0	38.0	33.1	30.7	26.2	(24)
(25)		10%	DB	26.5	26.9	24.8	32.5	41.4	53.0	57.3	53.4	43.4	35.6	32.3	26.0	(25)
(26)			MCWB	22.4	22.5	21.6	28.2	35.5	43.3	45.4	43.5	37.1	31.2	28.4	22.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.4	33.6	35.5	40.5	44.8	48.6	51.6	50.2	44.1	39.3	38.4	33.9	(27)
(28)			MCD B	37.9	41.2	40.9	49.1	52.2	54.0	55.9	52.9	47.5	44.1	43.1	38.5	(28)
(29)		2%	WB	29.6	27.8	29.8	35.5	41.2	47.0	49.9	48.5	41.2	35.8	34.1	30.6	(29)
(30)			MCD B	33.1	32.5	33.9	42.2	47.5	53.7	53.8	51.3	44.1	40.9	39.0	34.2	(30)
(31)		5%	WB	25.9	25.4	25.2	31.3	38.5	45.6	48.6	47.1	39.8	33.8	31.1	26.5	(31)
(32)			MCD B	30.2	30.6	28.5	36.7	44.0	52.5	53.1	50.5	43.2	38.3	35.6	30.0	(32)
(33)		10%	WB	22.6	22.2	21.6	28.0	36.4	44.3	47.6	46.1	38.6	31.7	28.3	23.0	(33)
(34)			MCD B	26.4	27.1	24.6	31.7	40.7	51.1	52.7	49.9	42.2	35.2	32.8	26.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	9.3	10.3	13.0	9.4	9.6	8.9	7.0	6.4	6.8	6.8	(35)	
(36)			MCD BR	12.7	13.1	14.3	26.8	15.3	14.1	13.8	12.9	9.6	10.2	11.6	10.6	(36)
(37)			MCWBR	9.5	9.4	10.5	10.6	8.0	5.4	5.1	5.8	5.9	6.6	8.5	8.7	(37)
(38)		5% WB	MCD BR	12.7	13.3	13.0	15.3	13.4	11.7	10.3	9.5	7.8	8.9	11.7	10.3	(38)
(39)			MCWBR	9.8	10.3	10.3	10.7	8.2	5.7	5.8	6.0	6.3	6.7	9.2	9.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	N/A	0.212	0.257	0.326	0.341	0.355	0.351	0.334	0.290	0.221	N/A	N/A	(40)	
(41)		taud	N/A	1.964	2.142	1.987	2.110	2.266	2.346	2.387	2.340	2.009	N/A	N/A	(41)	
(42)		Ebn at Noon	0	149	235	246	261	263	260	250	225	137	0	0	(42)	
(43)		Edh at Noon	0	12	23	38	39	35	31	26	20	11	0	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0	95	554	1285	1795	1937	1772	1174	600	172	0	0	(44)	
(45)		RadStd	0	12	41	102	117	131	162	102	48	14	0	0	0	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+1.57	+6.98	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page